Periodic GARCH: volatility with a calendar dummy

Does a shock after a weekend feed into tomorrow's risk as much as any other? Bollerslev and Ghysels (1996), reproduced.

Tested against	Bollerslev and Ghysels (1996), *Journal of Business & Economic Statistics*, Table 2 (p. 146): the 1,974 daily Deutsche mark / British pound returns of 1984–1991, with the paper's non-trading-day dummy (PollisDatasets `bg96`).
Missing in Julia	a dummy in the variance equation of a GARCH model, and the periodic GARCH model. ARCHModels.jl fits GARCH(1,1) only.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *Periodic GARCH: volatility with a calendar dummy* (`pgarch.ipynb`)
Result	32 of 42 printed values to every digit, most of the rest in the last digit. Plain GARCH agrees with ARCHModels.jl to about 10^{-13}.

The problem

Volatility is not the same every day. After a weekend or a holiday, news has piled up while markets were shut; a GARCH(1,1) treats that day like any other. Bollerslev and Ghysels (1996) let the model change with a calendar dummy $s(t)$: the mean, the level the variance returns to, and how much a shock feeds into the next day's variance can all differ when $s(t) = 1$. The idea carries over to any periodic calendar: days of the week, months, trading sessions.

The method

The mean is $\mu + \mu_1 s(t)$ and the conditional variance (p. 146, eq. 14 and the note to Table 2) is

$$\sigma_t^2 = \omega + \omega_1 s_t + (\alpha + \alpha_1 s_t)\,(\varepsilon_{t-1}^2 - \omega - \omega_1 s_{t-1}) + \beta\,(\sigma_{t-1}^2 - \omega - \omega_1 s_{t-1}).$$

$\omega + \omega_1 s(t)$ is the level the variance returns to, ω_1 the extra variance when $s(t) = 1$, and α_1 how much less, or more, a shock on such a day feeds forward. With $\alpha_1 = 0$ it is GARCH(1,1) with a dummy; with $\omega_1 = \alpha_1 = \mu_1 = 0$, plain GARCH(1,1). `pgarch` maximises the Gaussian likelihood and reports quasi-maximum likelihood (sandwich) standard errors; any parameter can be held at zero. `diagnostics` gives every statistic of Table 2: information criteria, skewness and kurtosis of the standardised residuals, the statistics of Ljung and Box (1978) on them and on their squares at 20 lags, and the paper's three loss functions.

What the tests show

Test	What is compared	Agreement
ARCHModels.jl	plain GARCH(1,1): estimates, robust s.e., log likelihood	to about 10^{-13}
Table 2, col. 5	GARCH(1,1) with the dummy: 19 values	16 to every digit, 3 in the last
Table 2, col. 6	P-GARCH(1,1): coefficients	every coefficient to 3 decimals
Table 2, col. 6	P-GARCH(1,1): s.e., SIC	4 in the last digit
Table 2, col. 6	skewness b_3, kurtosis b_4	$-.43$, 5.63 vs $-.46$, 5.87 (open)

The last-digit differences, side by side (Julia against the paper):

- Column 5: s.e. of the dummy .023 vs .022; b_3 $-.43$ vs $-.44$; b_4 5.88 vs 5.89.
- Column 6: s.e. of ω .171 vs .169, of ω_1 .027 vs .026, of α .042 vs .043; SIC -2212.8 vs -2212.9. These are what numerical Hessians of two programs give.

The economic result, reproduced

On ordinary days shocks persist almost indefinitely: $\alpha+\beta = 1.0006$, which rounds to 1.001 against the paper's 1.000. After a non-trading day they die out much faster: $\alpha+\alpha_1+\beta = .889$, as published. A Monday's shock feeds less into Tuesday's variance, so the periodic model's volatility jumps less after weekends than plain GARCH's; the notebook's example plots both over 1989–1990.

Still open

The skewness and kurtosis of column 6: $-.43$ and 5.63 here against $-.46$ and 5.87 in print, while the Q statistics, MSE, HMSE and log likelihood of the same standardised residuals all agree. The paper may compute b_3 and b_4 on a different sample

or residual definition. Not explained.

Further considerations

- **Several seasons.** The paper's general P-GARCH lets every parameter cycle with the calendar (five weekdays, twelve months). The notebook's single dummy is the case Table 2 tests; the general case needs one more index and no new ideas, and should come with a test of its own before it is offered.
- **Upstream.** A regressor in the variance equation is the missing piece in ARCHModels.jl; the dummy model is that plus a constraint. Column 5 is a ready-made test for such a contribution.
- **Asymmetry.** Calendar effects and leverage (EGARCH, GJR) are often studied together; a periodic GJR model would need a published table with its data before it becomes a new method here.

References

Bollerslev, T. and Ghysels, E. (1996). Periodic autoregressive conditional heteroscedasticity. *Journal of Business & Economic Statistics* 14(2), 139–151. doi:10.1080/07350015.1996.10524640

Ljung, G. M. and Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika* 65(2), 297–303.

Data: the returns as distributed with ARCHModels.jl, and the non-trading-day dummy as distributed with the R package rugarch (`dmbp`).

Methods written and tested with Claude Code (Anthropic) in Pollis, from the published papers and their data.
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